The AI-Detecting Eye: An Interactive Training Tool for Enhancing Human Detection of AI-Generated Content

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Background

The rapid advancement of generative AI has created a gap where, for many, AI-generated content is indistinguishable from authentic images. According to Lu et al. (2023), humans currently face a misclassification rate of 38.7% when evaluating state-of-the-art AI images. This percentage has likely only shrunk within the last three years. While humans are generally competent when it comes to identifying synthetic faces, they struggle significantly when it comes to urban and natural landscapes (Roca et al., 2025). The current methods to combat this issue often rely on automated detectors, but this does not empower the average person to detect AI-generated content on their own in the real world. This project aims to address the need for humans to have the ability to distinguish between reality and AI content on their own through an interactive website that teaches users to identify specific inconsistencies, artifacts, and traits left by AI generators.

Methodology

This research will utilize a quantitative approach to measure the efficacy of interactive training.

1. Data Acquisition: I will use the Fake2M and GenImage datasets to source 200 high-quality pairs of real and AI-generated images. I believe this is a better choice than manually generating 200 AI images, in terms of both efficiency and ethics.
2. Volunteer Recruitment: Forty participants will be recruited from the UCF student body via posters in the L3Harris Engineering Center and Engineering II building, in addition to announcements in undergraduate Computer Science lectures. As an incentive, each participant will receive a $10 Visa gift card upon completion of the study.

3. Application Development: A web-based application will be built that will feature a magnifying glass overlay that triggers when a user makes an identification, highlighting the specific spots where the AI failed. For example, it will specifically point out impossible geometry, lighting inconsistencies, etc.
4. Experiment Design: Participants will undergo a three-phase study:
    a. Pre-test: Detection accuracy will be measured.
    b. Training: Users will interact with the feedback tool.
    c. Post-test: A final evaluation will determine if overall detection skills improved.

Anticipated Outcomes

The primary result of this project will be a functional, open-source web game that is designed to test AI literacy. Statistically, I anticipate a significant improvement in participant accuracy after undergoing the website training. Based on similar structured training programs, I expect a p-value of < 0.05, indicating that interactive feedback is a mathematically effective method for increasing human skepticism and detection accuracy regarding AI-generated media (Samuel & Dev, 2025). I will present the project's conceptual framework and initial development progress at the UCF Showcase of Undergraduate Research Excellence (SURE) during Student Research Week which spans from March 29 to April 2, 2027. Once data analysis is completed in May, the full study results will be presented at the UCF Summer Undergraduate Research Showcase

Significance

The primary audience for this study are university students and social media users who are most frequently exposed to unverified synthetic media. This research will provide new insights into how humans detect synthetic content by determining which specific AI inconsistencies are the easiest and hardest for the human eye to learn. While previous studies have shown that lists of "tips" are ineffective, this project will provide insight into whether real-time, specific feedback creates a more lasting "mental model" for people. This insight can help developers design better browser extensions or social media filters that help users flag misinformation more quickly and efficiently.

References

Lu, Z., Huang, D., Bai, L., Qu, J., Wu, C., Liu, X., & Ouyang, W. (2023, December 15). *Seeing is not always believing: Benchmarking human and model perception of AI-generated images*. Advances in Neural Information Processing Systems. https://proceedings.neurips.cc/paper_files/paper/2023/hash/505df5ea30f630661074145149274af0-Abstract-Datasets_and_Benchmarks.html

Roca, T., Roman, A. C., Vega, J. T., Duarte, M., Wang, P., White, K., Misra, A., & Ferres, J. L. (2025, May 12). How good are humans at detecting AI-generated images? learnings from an experiment. arXiv.org. https://arxiv.org/abs/2507.18640

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Timeline

Dates	Tasks
January 1, 2027- January 25, 2027	Download and filter GenImage/Fake2M datasets; select 200 high-quality pairs.
January 26, 2027 - March 10, 2027	Develop frontend and implement the coordinate-based "Feedback Hotspots."
March 11, 2027 - April 15, 2027	Conduct user testing with volunteers.
April 16, 2027 - May 15, 2027	Analyze pre- and post-test data; finalize the technical research report.

Budget

Item	**Units and Cost**	**Total**
Interviewee incentives ($10 visa gift cards)	$10/person, 40 people	$400
Cloud Hosting (AWS)	$12/month, 5 months	$60
Domain Registration	$15/year, 1 year	$15
Promotion Materials	$25/poster, 4 units	$100
Total		**$575**